DAY 24 - ADVANCED

SQL CTEs & Window Functions

Primary tool: SQL

What you are learning

Readable multi-step queries, ranking and cumulative calculations.

Learning objectives

- Use WITH/CTE

- Apply ROW_NUMBER/RANK

- Create running totals

Worked practical

Rank performers within county and compute cumulative monthly actuals.

Source / Input	Method	Output / Check
Raw data	Apply the lesson skill	Result linked to source
Change one input	Recalculate / rerun	Confirm expected change
Control total / test	Validate	Explain the result

Guided practice

1. Recreate the lesson idea on a small example.
2. Change inputs and predict the result before recalculating.
3. Complete the supplied independent dataset in the required tool.
4. Create a control check or reconciliation.
5. Explain the result in ordinary language.

AI coaching

Ask AI to explain the difference between GROUP BY and window functions with a fresh example.

AI discipline

Attempt first. Use AI for explanation, hints, debugging, critique, test cases and mock questioning. Never submit output you cannot explain and independently verify.

Independent exercise

Create ranking and running-total queries.

Common failure modes

- Wrong grain or key.
- Formula/query/code answers a different question.
- Missing or unmatched records are silently ignored.
- Findings are not reconciled to the numbers.
- AI-generated logic is accepted without testing.

Mastery gate

[ ] I can reproduce the method without copying.
[ ] I can trace the result to the source.
[ ] I have a QA/control check.
[ ] I can explain a limitation.
[ ] I can answer an assessor's follow-up question without asking AI.